

Summary: *What Do Forward Interest and Exchange Rates Tell Us?*

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1 Paper Overview

1.1 Introduction & Context

Market participants, firms, and central banks all share a practical need: knowing what the market *as a whole* expects about the future path of interest rates and exchange rates. Forward rates—today’s contracted prices for transactions that will settle at a future date—are a natural candidate for reading those expectations. A widely invoked theory, the *expectations hypothesis*, holds that a forward rate equals the market’s average expectation of the corresponding future spot rate.

This paper investigates the empirical validity of that claim for New Zealand financial markets. Using one-month forward rates for (i) 90-day bank bill interest rates, (ii) the \$NZ/\$US exchange rate, (iii) the trade-weighted index (TWI), and (iv) the monetary conditions index (MCI) over January 1989 to March 1998, Ha and Reddell ask three questions: How accurately do forward rates predict future spot rates? Are there systematic patterns in the prediction errors? And do forward rates genuinely reflect market expectations? The broader motivation is to help the Reserve Bank understand what information forward prices convey for monetary-policy analysis.

1.2 Structure of the Paper

1. **Section 1 (Introduction):** Motivates the study and previews the main findings.
2. **Section 2 (What are forward rates and why are we interested in them?):** Defines forward contracts, explains hedging and speculation, and discusses the relevance of forward rates for central banks.
3. **Section 3 (How are forward rates determined?):** Presents the theoretical framework—arbitrage relationships (no-arbitrage for interest rates and covered interest parity for exchange rates) and the expectations hypotheses (the expectations hypothesis for interest rates and uncovered interest parity for exchange rates).
4. **Section 4 (What does the evidence show?):** Reports empirical results: forecast accuracy of forward rates, systematic patterns in forecast errors, the role of risk premia and inflation expectations, comparison with survey data, and a discussion of why NZ interest rates are persistently higher than US rates.
5. **Section 5 (Conclusion):** Summarises findings and policy implications.

1.3 Conclusions

The paper reaches four principal conclusions:

1. Forward rates are poor predictors of the *level* of future spot rates over a one-month horizon. Directional accuracy is statistically no better than a coin toss, and explanatory power is below 6% for all instruments studied.
2. Forward rates have *systematically* under-estimated actual changes in spot rates, with the sign of the error tracking the cyclical trend in each series. The authors attribute this largely to persistent errors in forecasting inflation and hence monetary policy.
3. Forward **interest rates** and forward **MCIs** are nonetheless reasonable proxies for average market expectations, as confirmed by comparison with Reserve Bank survey data.
4. Forward **exchange rates** do *not* reflect market expectations. They have priced in a NZ dollar depreciation in virtually every month of the sample, driven by a currency-related risk premium that keeps NZ interest rates above US rates.

1.4 Key Takeaways

1. **Forward rates $\neq$ forecasts.** One-month forward rates carry almost no predictive power for the level or even the direction of future spot rates. Their value lies elsewhere.
2. **Systematic forecast errors track cycles.** Forward rates persistently under-predict changes in spot rates, and this pattern mirrors the trend direction of each price series—a phenomenon linked to slow learning about inflation dynamics.
3. **Forward interest rates $\approx$ expectations; forward exchange rates $\neq$ expectations.** Interest-rate forwards line up well with survey data, but exchange-rate forwards embed a persistent currency risk premium rather than pure expectations.
4. **The risk-premium wedge reflects global capital-market structure.** New Zealand's higher interest rates (and the implied depreciation in forward exchange rates) stem from a general preference for US-dollar assets and NZ-specific factors such as high net foreign debt, not from default risk.
5. **UIP is useful *ex ante*, not *ex post*.** Uncovered interest parity describes how markets *react* to surprises (e.g. unexpected monetary tightening) even though it fails as an average predictor of realised exchange-rate movements.

2 Key Points

2.1 Definition and Purpose of Forward Contracts

A **forward contract** is an agreement between two parties today to buy or sell an asset at a specified future date, at a price agreed upon today (the *forward rate*). In the NZ context the paper examines two types:

- **Forward exchange rate:** e.g. a one-month \$NZ/\$US forward rate of 0.5000 means a customer can exchange NZ dollars for US dollars at 0.5000 in one month, regardless of the prevailing spot rate.
- **Forward interest rate:** e.g. a one-month forward 90-day rate of 8% means a customer can borrow or lend for 90 days, starting in one month, at 8%.

The primary economic function of forward contracts is **hedging**—allowing exporters, importers, borrowers, and investors to eliminate price uncertainty. Banks provide these products and also use them to manage their own exposures. Forward TWI and MCI contracts, while available, are mainly used by institutional investors to speculate on the future stance of monetary policy rather than to hedge real exposures.

Central banks are interested in forward rates because, if they approximate market expectations, they offer a window into what the market collectively anticipates about future policy actions (see [Key Point 2.3](#)).

2.2 Arbitrage and Forward Interest Rates

Arbitrage ensures a precise no-arbitrage link between current spot interest rates and forward interest rates. In the paper's example: if the one-year rate is 9% and the current six-month rate is 8%, arbitrage forces the six-month forward interest rate to 10%, so that

$$\frac{6}{12} \times 8\% + \frac{6}{12} \times 10\% = 9\%.$$

Any deviation from this relationship creates a riskless profit opportunity that the market quickly eliminates. This relationship is mechanical and always holds in liquid markets; it pins down the *level* of forward interest rates given the term structure, but it does not, by itself, tell us what the forward rate *means* in terms of expectations (that interpretation comes from the expectations hypothesis discussed in [Key Point 2.3](#)).

2.3 The Expectations Hypothesis for Interest Rates

The **expectations hypothesis** (EH) posits that forward interest rates reflect the market's average expectation of future spot interest rates. In the example from [Key Point 2.2](#), a six-month forward rate of 10% implies that the market expects the six-month spot rate to be 10% in six months' time.

The mechanism works through demand and supply: if investors expect the future rate to be *lower* than the forward rate, they prefer long-term instruments, bidding down long-term rates until the implied forward rate aligns with expectations. Conversely, if they expect higher future rates, demand shifts to short-term instruments.

The EH is qualified by the possibility of a **risk premium**: investors who roll over short-term loans face the risk that rates will move against them, so risk-averse agents may accept a wedge between the forward rate and the expected future rate. The empirical relevance of such a premium at one-month horizons is examined in [Key Point 2.7](#).

2.4 Covered Interest Parity (CIP) and Forward Exchange Rates

Covered interest parity (CIP) is the exchange-rate counterpart of the interest-rate no-arbitrage condition. CIP states:

$$r_t^{US} - r_t^{NZ} = f_t - s_t$$

where r_t^{US} and r_t^{NZ} are the relevant interest rates, f_t is the (log) forward exchange rate, and s_t is the (log) spot exchange rate. This means the forward exchange rate prices in a depreciation (or appreciation) of the NZ dollar exactly equal to the interest-rate differential.

CIP is enforced by arbitrage: if the forward rate did not fully reflect the interest differential, investors could lock in a riskless excess return by borrowing in the low-rate currency, investing in the high-rate currency, and covering exchange-rate risk with a forward contract. As a result, **interest rate differentials are almost always a very good guide to where forward exchange rates are in the market**, a fact the paper emphasises repeatedly.

2.5 Uncovered Interest Parity (UIP)

Uncovered interest parity (UIP) extends the logic of CIP to form a hypothesis about exchange-rate *expectations*:

$$r_t^{US} - r_t^{NZ} = E_t s_{t+1} - s_t$$

UIP asserts that the interest-rate differential equals the market’s *expected* depreciation of the NZ dollar. Unlike CIP, UIP is *not* enforced by arbitrage because the investor remains exposed to exchange-rate risk (an “uncovered” position).

The paper stresses that interest rates and exchange rates are determined *simultaneously*: UIP is as much a theory of interest-rate determination as of exchange-rate expectations. Empirically, UIP in its strict form fails—as discussed in [Key Point 2.9](#)—but it remains useful as an *ex-ante* framework for analysing market reactions to surprises ([Key Point 2.10](#)).

2.6 Forecast Accuracy of Forward Rates

Table 1 reproduces the paper’s key summary statistics. The evidence is stark: forward rates are poor predictors of future spot rates at a one-month horizon.

Table 1: Forward rate “forecast errors”: summary statistics (Jan 1989 – Mar 1998).

	90-day rate (basis pts)	\$NZ/\$US (%)	TWI (%)	MCI (basis pts)
Average (absolute) difference	37	1.4	1.2	61
Standard deviation	50	1.9	1.5	82
Smallest difference	0	0	0	1
Largest difference	153	6.8	4.8	261
Picking the direction	56%	49%	44%	57%
Explanatory power (R^2)	0.7%	0.9%	0.8%	6%

Key observations:

- The average absolute “forecast error” for 90-day interest rates is 37 basis points, and roughly 95% of errors fall within ± 100 basis points.
- Directional accuracy (picking whether rates rise or fall) is not significantly different from 50% in any case—no better than a coin flip.
- The R^2 from regressing actual changes on forward-implied changes is below 6% for all instruments, meaning forward rates explain almost none of the variation in actual changes.

The paper argues this is unsurprising: financial prices are volatile and short-term movements are dominated by information that was not, and often could not have been, known at the time the forward rate was set (see [Key Point 2.7](#) for the systematic component of these errors).

2.7 Systematic Patterns in Forecast Errors

Despite the overall inaccuracy, the forward-rate forecast errors display a striking *systematic* pattern. The paper plots 12-month moving averages of forecast errors alongside the level of each spot rate and finds:

- When spot rates were **trending upward**, forward rates persistently *under-estimated* the rise, producing **positive** forecast errors.
- When spot rates were **trending downward**, forward rates persistently *under-estimated* the fall, producing **negative** forecast errors.

This cyclical correlation implies that simple trend-following trading rules could, in hindsight, have generated excess returns—for example, buying NZ-dollar forward contracts during the 1993–1996 appreciation. However, these strategies are inherently risky because they require correctly identifying the prevailing trend in real time and tolerating substantial short-term volatility.

This pattern is **not explained by a stable insurance-type risk premium** ([Key Point 2.8](#)). The authors instead attribute it to persistent inflation forecast errors ([Logical Chain 3.1](#)).

2.8 Ruling Out the Insurance Risk Premium Explanation

Can the systematic forecast errors be explained by a risk premium embedded in forward rates? The paper considers this carefully and concludes that a conventional insurance-type risk premium does not fit the data, for three reasons:

1. A risk premium should produce a *relatively stable* margin between forward and future spot rates. Instead, the margin changes sign with the trend direction.
2. To generate the observed pattern, the premium would need to be *negative* when rates are trending up and *positive* when trending down—a requirement without obvious economic motivation.
3. Financial markets are two-sided: risk-averse borrowers would pay a positive premium for certainty, but risk-averse lenders would accept a negative premium. These forces should roughly cancel at a one-month horizon.

This motivates the alternative explanation based on inflation expectations discussed in [Key Point 2.7](#) and formalised in [Logical Chain 3.1](#).

2.9 Forward Rates vs. Survey Data: Interest Rates and MCIs

To assess whether forward rates really capture market expectations (as distinct from realised outcomes), the paper compares forward-implied expected changes with the Reserve Bank’s *Survey of Expectations*, which polls over 100 businesses and market participants each quarter.

Forward 90-day interest rates align well with surveyed expectations: the two series are similar in both direction and magnitude. This suggests that, despite poor ex-post accuracy ([Key Point 2.6](#)), forward interest rates *are* a reasonable guide to what the market expected at the time.

Forward MCIs also track the composite survey MCI expectation reasonably well, which the authors note indicates that the MCI forward is dominated by the interest-rate component.

2.10 Forward Exchange Rates Do Not Reflect Expectations

In sharp contrast to interest rates, **forward exchange rates do not look like market expectations**. Two pieces of evidence:

1. The forward \$NZ/\$US rate has priced in a NZ dollar depreciation in *every single month* of the nine-year sample—even during the sustained appreciation of 1993–1996. It is implausible that the market expected depreciation every month for nine years.
2. The forward-implied expected change diverges substantially from surveyed exchange-rate expectations, particularly between 1992 and 1995, when surveys anticipated appreciation while forwards continued to imply depreciation.

The explanation lies in CIP ([Key Point 2.4](#)): because NZ interest rates are persistently above US rates, CIP mechanically produces a forward rate below the spot rate, i.e. an implied depreciation. But that implied depreciation reflects the *interest differential*, not genuine expectations about the exchange rate's direction. The interest differential itself is driven by a **currency risk premium** ([Key Point 2.11](#)).

2.11 The Currency Risk Premium and NZ–US Interest Differentials

The persistent NZ–US interest differential (averaging 3.6 percentage points per year over the sample) cannot be attributed to default risk: NZ Government \$US-denominated bonds trade at only about 15 basis points above US Treasuries, consistent with NZ's AA+ vs. AAA credit rating.

Instead, the paper identifies a **currency-related risk premium** arising from two sources:

1. **Global preference for US-dollar assets.** The US sits at the centre of the world's financial system; central-bank reserves are overwhelmingly held in USD, JPY, and DEM; US financial markets are the deepest and most liquid; and US investment funds have historically had a strong home bias. These structural features mean that investors require compensation to hold non-USD assets.
2. **NZ-specific factors.** New Zealand has the highest ratio of net foreign debt to GDP among OECD economies. Most overseas investors in NZ-dollar bonds hold unhedged positions, so their returns depend on exchange-rate movements. The limited appetite for that additional risk means NZ must offer higher interest rates to attract capital.

The implication is that **UIP does not hold in its strict form**: short-term exchange-rate expectations are not fully reflected in interest-rate differentials because the differential includes a risk-premium component. This is a finding shared by studies across many countries.

2.12 UIP as an Ex-Ante Framework

Although UIP fails as an average ex-post predictor (forward exchange rates do not equal future spot rates on average), the paper argues it remains a powerful *ex-ante* framework for understanding how markets respond to new information.

The key example is New Zealand's experience over 1995–1996, when inflation repeatedly surprised on the upside. Each unexpected tightening of monetary policy produced:

1. A rise in NZ interest rates.
2. An immediate, discrete jump (“step change”) in the exchange rate.
3. A relatively flat exchange rate thereafter—consistent with the market expecting a subsequent depreciation to offset the wider interest differential.

This reaction pattern—an immediate exchange-rate appreciation followed by expected depreciation—is exactly what UIP predicts. The NZ dollar did not actually depreciate during this period, but only because each surprise tightening re-set expectations before the anticipated depreciation could occur. In the absence of repeated positive inflation surprises, the depreciation would likely have materialised. Hence, UIP describes how expectations *adjust* to surprises, even if it cannot explain realised outcomes over the full sample.

3 Logical Chains

3.1 From Inflation Forecast Errors to Systematic Under-Prediction of Spot Rate Changes

This chain explains the systematic pattern documented in [Key Point 2.7](#): why forward rates persistently under-estimated the magnitude of actual changes in future spot rates.

Step 1. Inflation surprises occur persistently in one direction: When inflation is rising, both the central bank and financial markets repeatedly under-estimate the strength of inflationary pressures.

Explanation: This “slow learning” about the inflation process means that each new data release reveals stronger-than-expected inflation, period after period.

Step 2. Monetary policy tightens by more than expected: Because inflation is persistently under-predicted, the central bank raises interest rates more aggressively than the market anticipated.

Explanation: Monetary-policy decisions respond to actual inflation outcomes, so under-estimated inflation leads to larger-than-expected tightening.

Step 3. Short-term interest rates and the exchange rate rise by more than expected: Higher-than-anticipated policy rates push both the short end of the yield curve and the exchange rate above forward-implied levels.

Explanation: The exchange-rate response is consistent with UIP ([Key Point 2.5](#)): unexpected tightening causes an immediate appreciation.

Step 4. Forward rates systematically under-predict the upward changes: Because the market consistently expected less tightening than materialised, forward rates were set too low relative to the realised spot rates, producing positive forecast errors during tightening cycles.

Explanation: This follows directly from Steps 1–3 applied repeatedly.

Step 5. The symmetric pattern holds in easing cycles: When inflation begins to fall, the same slow-learning dynamic operates in reverse—inflation falls faster than expected, policy eases more than anticipated, and spot rates drop below forward-implied levels, producing negative forecast errors.

Explanation: The mechanism is identical with reversed signs.

Conclusion. Persistent inflation forecast errors propagate into systematic, cycle-correlated under-prediction of spot-rate changes by forward rates.

Explanation: The root cause is not a failure of the forward market itself, but a failure—shared by central banks and markets—to anticipate inflation dynamics. Because both inflation and monetary policy exhibit persistence, errors in one period carry into the next, generating the cyclical pattern visible in the 12-month moving averages of forecast errors.

3.2 From CIP and the Risk Premium to the Persistent Depreciation Bias in Forward Exchange Rates

This chain explains why forward exchange rates have always implied NZ dollar depreciation, as documented in [Key Point 2.10](#).

Step 1. CIP holds: Arbitrage ensures that the forward exchange rate reflects the interest-rate differential between NZ and the US ([Key Point 2.4](#)).

Explanation: Any deviation from CIP would create riskless arbitrage profits that are quickly exploited.

Step 2. NZ interest rates are persistently higher than US interest rates: Over the sample, one-month NZ rates exceeded US rates by an average of 3.6 percentage points per year ([Key Point 2.11](#)).

Explanation: This differential is driven by a currency risk premium—global preference for USD assets and NZ-specific factors such as high net foreign debt.

Step 3. CIP translates the positive interest differential into a forward rate below the spot rate: If $r^{NZ} > r^{US}$, then CIP ($f_t - s_t = r_t^{US} - r_t^{NZ}$) implies $f_t < s_t$, i.e. the forward rate prices in NZ dollar depreciation.

Explanation: This is a mechanical implication of the CIP formula and the sign of the interest differential.

Step 4. Because the differential persists, the implied depreciation appears in every month of the sample: The forward exchange rate has never priced in NZ dollar appreciation.

Explanation: The structural risk premium ensures that NZ rates remain above US rates throughout the sample.

Conclusion. Forward exchange rates embed a persistent depreciation bias that reflects the currency risk premium in NZ interest rates, not genuine market expectations of exchange-rate depreciation.

Explanation: Since CIP is enforced by arbitrage and the interest differential contains a risk-premium component, forward exchange rates mechanically inherit that premium and cannot be read as pure expectations—a conclusion confirmed by the divergence between forwards and survey data ([Key Point 2.10](#)).

4 Key Terms & Definitions

Arbitrage

The near-instantaneous buying and selling of the same asset across markets to exploit price discrepancies, yielding a riskless return. Arbitrage involves no open position and ensures price consistency across markets.

Basis point (bp)

One hundredth of a percentage point, i.e. $1 \text{ bp} = 0.01\%$. Thus $100 \text{ bp} = 1\%$.

Covered Interest Parity (CIP)

The no-arbitrage condition for exchange rates: $r_t^{US} - r_t^{NZ} = f_t - s_t$. It states that the forward exchange rate must offset the interest-rate differential so that a hedged cross-border investment earns the same as a domestic investment. CIP holds reliably in practice because deviations create riskless profit opportunities.

Currency risk premium

An additional return that investors require to hold assets denominated in a currency other than the US dollar. In the paper, this premium explains why NZ interest rates exceed US rates by more than default risk alone would justify.

Expectations Hypothesis (EH)

The theory that forward interest rates equal the market's average expectation of future spot interest rates. It is the interest-rate analogue of UIP.

Explanatory power (R^2)

The share of variance in actual spot-rate changes that is explained by changes implied by forward rates, as estimated by an OLS regression. In the paper, R^2 values are uniformly very low ($< 6\%$).

Forecast error

Defined in the paper as the actual future spot rate minus the previous month's forward rate, i.e. $\text{error}_t = s_{t+1} - f_t$. Positive values mean the spot rate turned out higher than the forward rate implied.

Forward contract / Forward rate

An agreement to transact at a future date at a price (the forward rate) fixed today. The paper studies one-month forward rates for 90-day interest rates, the \$NZ/\$US exchange rate, the TWI, and the MCI.

Monetary Conditions Index (MCI)

A weighted combination of 90-day interest rates and the TWI, used by the Reserve Bank of New Zealand as an approximate summary measure of the overall stance of monetary conditions. The December 1996 quarter average is normalised to 1000.

Open position

An unhedged financial exposure. A speculator who buys forward NZ dollars without an offsetting obligation holds an open position in the NZ dollar and bears exchange-rate risk.

Risk premium (insurance-type)

A price that risk-averse agents are willing to pay to eliminate uncertainty via a forward contract. The paper argues that a stable insurance-type risk premium does *not* explain the systematic forecast errors observed in the data.

Speculation

Taking an open position based on an opinion about future prices. Unlike arbitrage, speculation is inherently risky; actual returns depend on the realised future spot rate.

Spot rate

The current market price for immediate (or near-immediate) transactions. For foreign exchange, “spot” technically refers to settlement in two business days.

Trade-Weighted Index (TWI)

A weighted average of the NZ dollar exchange rate against the currencies of New Zealand’s five largest trading partners (US, Australia, Japan, UK, Germany as of 1998).

Unbiasedness hypothesis

The proposition that forward rates are, on average over time, equal to the future spot rates they correspond to—i.e. that forward rate forecast errors have mean zero. Formally tested in the literature but largely unresolved, especially for exchange rates.

Uncovered Interest Parity (UIP)

The hypothesis that the interest-rate differential between two countries equals the market’s expected rate of exchange-rate depreciation: $r_t^{US} - r_t^{NZ} = E_t s_{t+1} - s_t$. Unlike CIP, UIP is not enforced by arbitrage and fails empirically in the strict sense, though it provides a useful framework for understanding market reactions to monetary-policy surprises.